

Following this session, students should be able to explain the mechanisms of action, resistance, general adverse effects, dose-limiting toxicities, cancer uses, and drug interactions:

1. Explain the fundamental principles of nonselective cytotoxic chemotherapy on inducing DNA damage in relation to cell cycle phases and kinetics.
2. Relate the anticancer mechanisms actions of each category of antineoplastic drugs to their effects.
3. Correlate the mechanisms of resistance with the mechanisms of drug action and cellular mechanisms of detoxification and reduced drug accumulation.
4. Identify the dose-limiting toxicities of the cytotoxic chemotherapies.
5. Apply the mechanistic and pharmacokinetic properties of the anticancer drugs to their potential adverse effects and drug interactions when presented with a case vignette.
6. Create a concept map or other technique integrating the mechanisms of action and resistance, potential toxicities, drug-disease and drug-drug interactions to the selection of drugs for combination antineoplastic therapeutic regimens.

Chemotherapy:

Cytotoxic chemicals that kill cells or stop cell division leading to apoptosis

when proteins that initiate apoptosis, such as p53, are functional.

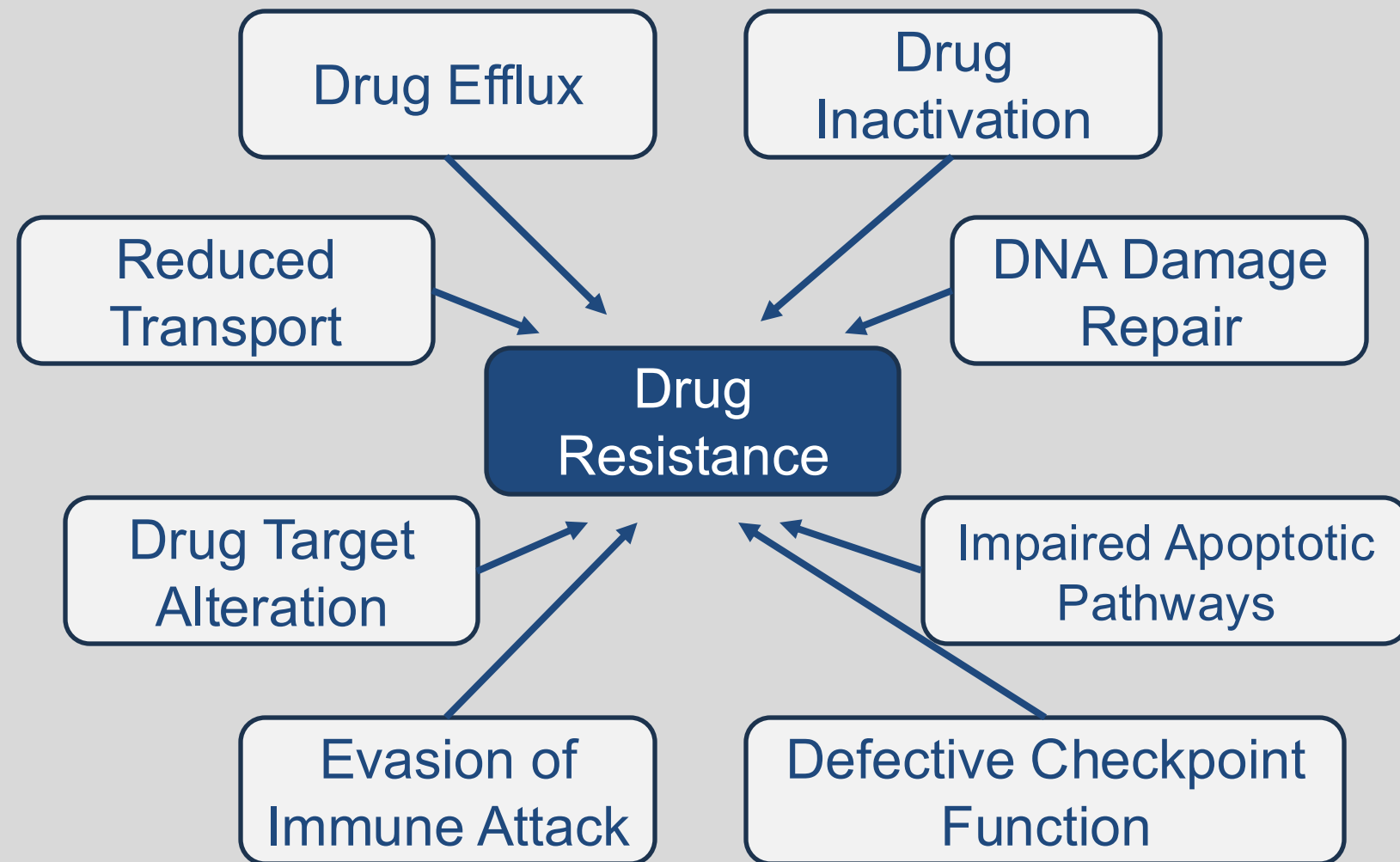
Alkylating agents All phases of cell cycle	Nucleophilic attack on DNA → DNA cross-linking → DNA damage → activate DNA mismatch repair (MMR) pathway → DNA strand breaks by repair enzymes → apoptosis.
Platinum analogs All phases of cell cycle	Form covalent metal adducts with DNA → DNA cross-linking → DNA damage → activates MMR ultimately leading to apoptosis
Antimetabolites S phase phase	Folate, purine, or pyrimidine analogs that inhibit DNA synthesis and repair in proliferating cells → p53 activation → apoptosis.
Antitumor antibiotics G2/M phase primarily	Intercalate in DNA, inhibit topoisomerase II, and generates free radicals → irreparable strand breaks, oxidative damage to DNA, proteins, and cell membranes.
Antimitotics M phase	Inhibit polymerization in microtubules (vinca alkaloids) or stabilize microtubules and prevent depolymerization (taxanes) → suppress spindle microtubule function → cell cycle arrest → apoptosis
Camptothecin analogs S phase	Form drug-DNA-topoisomerase I complex → prevents resealing step → DNA damage accumulates → apoptosis
Epipodophyllotoxins late S and G2 phase	Form drug-DNA-topoisomerase II complex → irreparable strand breaks → DNA damage accumulates → apoptosis

- **What are the goals of cancer therapy?**
 - To rid the body of malignant cells
 - If cure is not possible, then the goal is to shrink the tumors and relieve symptoms.
- **Antineoplastic therapy entails surgery, radiation, anticancer drugs**

What are the patient's goals?

Mechanisms of Resistance

1. Increased rate of DNA repair
2. Defective cell cycle checkpoint function or mismatch repair (MMR) and DNA damage repair systems
3. Impaired apoptotic pathways
4. Decreased drug accumulation:
 - Expression of efflux pumps
 - Reduced transport into cancer cell
 - Inactivation of drugs
 - Decreased activation of prodrugs
5. Increased expression of drug detoxifying proteins, such as glutathione conjugation, that neutralize free radicals
6. Target alterations with decreased affinity for the drugs
7. Tumor evasion of immune attack



Chemotherapy Treatment Strategies

1. Drugs for slow-growing cancers and solid tumors damage DNA.
2. Drugs for rapidly proliferating tumors interfere with DNA synthesis or mitosis.
3. Endocrine therapy for hormone-responsive tumors lowers the amount or interferes with the actions of the hormone, eg, estrogen or testosterone.
4. Multi-drug combinations include cytotoxic drugs acting at specific stages of cell cycle and/or targeted agents that inhibit cancer growth pathways or boost the immune response against the tumor.
 - To maximize the effect and for synergy
 - To minimize the development of resistance
5. Maximum individual doses are administered as frequently as possible, depending on:
 - The time needed for the body to recover from side effects, such as neutropenia
 - The type and stage of the cancer
 - The treatment goals, such as curative or palliative
6. Avoid overlap in major toxicities (ideally)
7. Manage toxicities → antiemetics; analgesics; etc.; non-medical strategies

Toxicities of Antineoplastic Agents

Chemotherapies disrupt DNA function in proliferating normal cells along with tumor cells.

A caution for all treatment regimens:

The pharmacokinetics and toxicities of cancer drugs vary amongst individual patients.

It is *imperative* to:

- recognize toxicities early,
- alter doses or discontinue offending medication to relieve symptoms and reduce risk, and
- provide vigorous supportive care.

Dose-limiting toxicities are specific adverse effects of a drug severe enough to prevent an increase in the dose.

They are critical in determining the maximum tolerated dose during clinical trials, especially in oncology.

Rapidly proliferating cells:

- Bone marrow suppression: ↑ susceptibility to **infection; bleeding**
- GI: Intestinal epithelium: Severe vomiting, mucositis, stomatitis
- Alopecia
- Cutaneous reactions

Tumor lysis syndrome:

- Metabolic complication of chemotherapy,
- Usually in lymphomas and leukemias

Drug-specific toxicities, acute and delayed

- Cardiac; Pulmonary; Hepatic; Renal; Neuro

Secondary malignancy:

- Leukemia and other malignancies
- Teratogenicity / Fetotoxicity
- Cancer immunotherapy: See Part 2

Alkylation of DNA: Nucleophilic attack

Cell death depends on cancer cell
recognition of the adduct



**activation of DNA mismatch repair
(MMR) pathway creates DNA
strand breaks by repair enzymes**

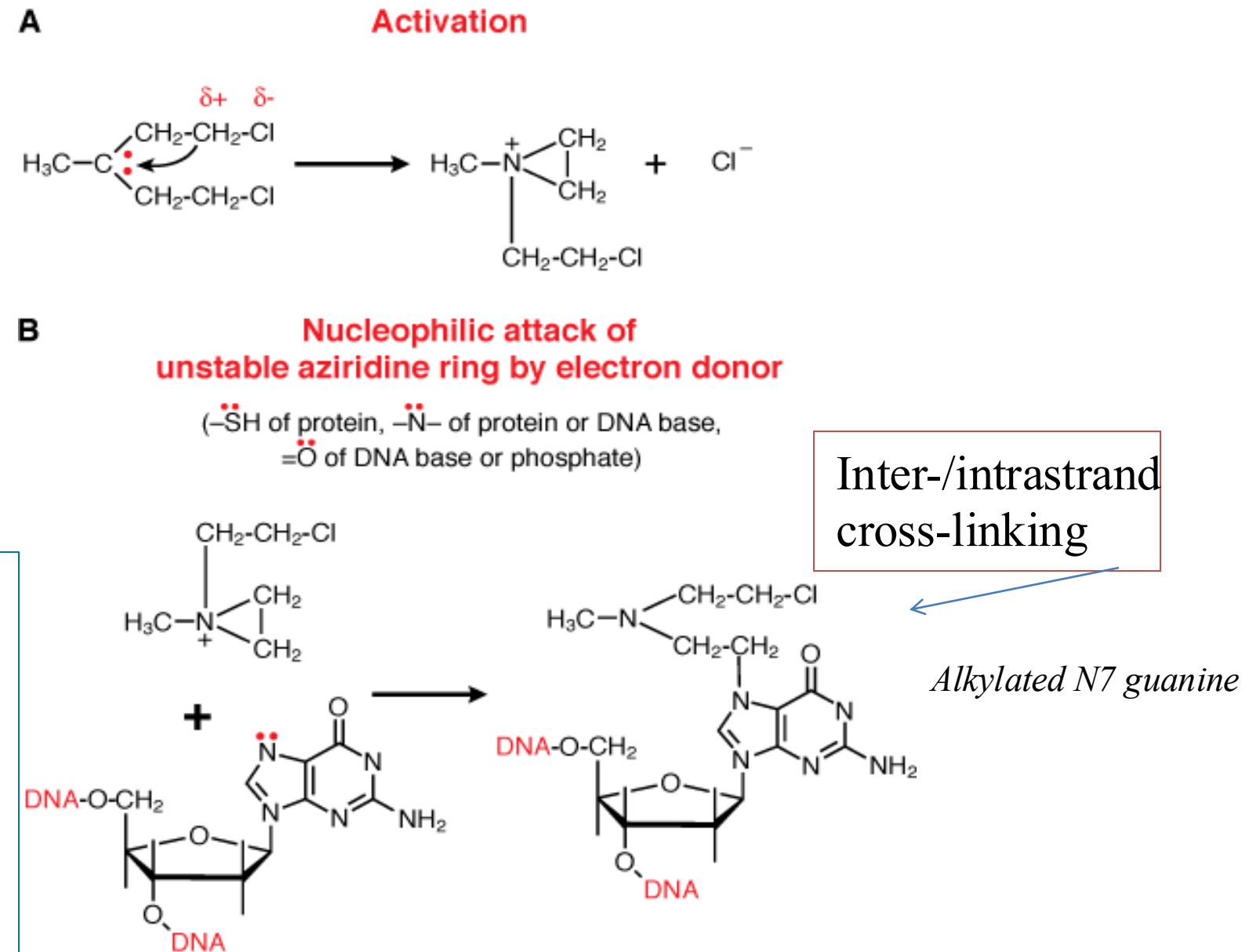


triggers apoptotic response

Essential mechanisms for cell death:

- 1) Recognition of the adduct
- 2) Activation of checkpoint proteins
- 3) Intact MMR / DDR systems, which induce DNA strand breaks by repair enzymes
- 4) Normal genes for p53 and other proteins that trigger apoptosis

**Resistance develops rapidly when a drug is
used as a single agent.**



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*:
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Toxicities of alkylating agents are related to DNA damage of normal rapidly dividing cells and can be quite severe.

Rapidly proliferating cells:

- Bone marrow suppression: DOSE-LIMITING **anemia, infection; bleeding**
- GI mucosal toxicity: **Severe** vomiting, mucositis, stomatitis
- Cutaneous reactions / Hair loss (alopecia)

Tumor lysis syndrome in the treatment of lymphoma and leukemia

- Rapidly lysing cells release contents into blood stream:
- Hyperuricemia, hyperphosphatemia, hyperkalemia, hypocalcemia,
- Acute kidney injury

Organ toxicity

- Hepatic sinusoidal obstruction syndrome
- Pulmonary – “Busulfan lung”
- Peripheral neuropathy
- Extravasation (leakage) from injection site: vesicant (causes blisters / ulcerations)

Secondary malignancy: Leukemia

- **Pregnancy:** Teratogenic
- **Reproductive system:** amenorrhea | azospermia

Check your knowledge

What class of drugs induces cell death by nucleophilic attack on DNA bases, esp. the N7 guanine, leading to inter- and intrastrand cross-linking, abnormal base pairing, miscoding, and strand breakage.

What cellular mechanisms are essential for cell death?

What is the active cytotoxic metabolite of cyclophosphamide (and ifosfamide)?

Check your knowledge

What is the name of the urotoxic metabolite and how can the toxicity be mitigated?

What is a mechanisms of tumor resistance to cyclophosphamide and ifosphamide?

Cyclophosphamide is used in the treatment of what categories of cancers?

List the common and serious toxicities of cyclophosphamide.

Resistance to Platinum Analogs

Decreased uptake	Efflux	Inactivation	Repair
CTR1 copper transporter 1. Down-regulation, and 2. ↓ expression Decreased uptake is the predominant mechanism of resistance	ATP7A and ATP7B copper transporters Physiologic function: removal of excess copper from the cell	↑ intracellular levels of 1. Glutathione, 2. Metallothionein, 3. Other sulfhydryls These molecules bind to and inactivate the drugs.	Overexpression of: 1. Nucleotide excision repair components, 2. Base-excision repair, 3. Mismatch repair (MMR), 4. double-strand-repair breaks

Cross-resistance may occur between cisplatin and carboplatin
Tumors resistant by MMR are not cross-resistant to oxaliplatin

Platinum Analogs

Common features	IV, poor CNS penetration renal clearance dosage adjustment in renal impairment	Synergy with radiation therapy Interaction: T before P Administer taxane before platinum to limit myelosuppression and increase efficacy
Cisplatin	Covalent binding to plasma proteins Nonenzymatic inactivation (unbound fraction) Vigorous saline hydration $t_{1/2\text{terminal}}$ up to 5 days	Major antitumor activity in a broad range of solid tumors
Carboplatin	Less reactive Less renal and GI toxicity and vigorous IV hydration is not required (but recommended)	Broad-spectrum activity against a wide range of solid tumors
Oxaliplatin	Highly reactive Cytotoxicity does not depend on active mismatch repair system for efficacy. (Bulky ligands create a different distortion pattern of DNA helix.)	Colorectal and gastric cancer Oxaliplatin is a component of first-line chemotherapy of advanced colorectal cancer (FOLFOX)

Dose-limiting toxicities (in red) and other major toxicities

Reminder: Being aware of major toxicities is important for optimizing the patient's therapy. Patients should be monitored for them, and the toxicities should be mitigated, if possible.

Cisplatin

Highly emetogenic – Vomiting: persistent, acute and delayed neurotoxicity; ototoxicity; myelosuppression, peripheral neuropathy

Nephrotoxicity vigorous chloride diuresis with normal saline + diuretic to reduce toxicity

Carboplatin

More tolerable – Toxicities of cisplatin but less severe.

Myelosuppression

Oxaliplatin

Hypersensitivity / Anaphylaxis – can occur within minutes
Toxicities similar to cisplatin but less severe

Peripheral neuropathy acute (exacerbated by cold) and delayed progressive neuropathy

All: Leukemia, pulmonary fibrosis months to years later

Check your knowledge

How does the mechanism of the platinum analogs compare with the mechanism of the alkylating agents?

What are the main mechanisms of resistance to platinum analogs?

Check your knowledge

What is the definition of dose-limiting toxicities?

What are the dose-limiting toxicities of cisplatin, carboplatin, and oxaliplatin?

In sequential administration of a platinum analog with paclitaxel or docetaxel, why is the taxane administered first?

Antimetabolites Class Toxicities – Acute and Delayed

Like the alkylating agents, antimetabolites can cause:

- Bone marrow suppression – anemia, neutropenia, thrombocytopenia
- GI toxicity – nausea, vomiting, diarrhea, mucositis, stomatitis
- Hepatotoxicity – ranging from elevated serum transaminases to liver failure
- Pulmonary toxicity – pneumonitis, interstitial fibrosis
- Nephrotoxicity – impaired kidney function and interstitial toxicity as drug deposits in the kidney parenchyma and tubular epithelial cells
- Neurotoxicity – peripheral neuropathy
- Cutaneous reactions – rash, hand-foot syndrome, alopecia

Methotrexate (MTX) Pharmacokinetic Properties

Property	Comment
A IV administration for cancer treatment; intrathecal (IT) for ALL	Oral, IM, subQ for weekly nononcologic immunosuppressant indications
D Taken up by reduced folate transporter ~50% plasma protein bound Narrow therapeutic window Distributes in body water Poor CNS penetration Sustained concentrations in liver/kidney	Drugs that displace methotrexate from albumin → ↑ risk of severe myelosuppression MTX is a Substrate of BCRP, P-gp, OATP (liver) BCRP: breast cancer resistance protein, an efflux transporter expressed in brain, GI tract, placenta, and testes and eliminates substrates from liver via biliary tract and via transport into the proximal tubule.
M Intracellular polyglutamation Low intestinal and hepatic metabolism	Active MTX-PG accumulates intracellularly and is slowly eliminated (ion trapping inside cell)
E Urine, 80%-90% unchanged drug	Dose adjustments for renal impairment
t _{1/2} Age and dose-dependent	Children: 1-6 h (ALL treatment) Adults: 8-15 h (cancer treatment)

Methotrexate has many therapeutic uses.

Oncology uses: **Broad spectrum:** solid and liquid tumors, including childhood ALL

Autoimmune disease: Weekly low dose for immunosuppressive effects

Abortifacient: Tubular ectopic pregnancy

Methotrexate drug-specific toxicities, in addition to class toxicities

Bone marrow suppression

Mucositis, stomatitis, nausea, vomiting, diarrhea

Hepatotoxicity is cumulative and progressive

Pneumonitis can occur at any time

Neurotoxicity with high IV dose:

- paralysis, palsy, seizure, coma
- arachnoiditis with intrathecal admin
- demyelinating encephalitis with radiation

Pregnancy: Fetal loss - abortifacient

Reproductive toxicities

Impaired fertility- defective oogenesis / spermatogenesis

Leucovorin rescue: Converted to active methylene-FH₄ → “rescue” host cells from:
folinic acid Bone marrow toxicity, hepatotoxicity, mucositis/stomatitis

Leucovorin does not reverse neurotoxicity or pulmonary toxicity.

5-FLUOROURACIL

Activation	Enzymatic ribosylation and phosphorylation of 5-FU to the active nucleotides F-UMP and F-dUMP
Distribution	Throughout body, including CSF and brain tissue
Metabolism	5-FU dihydropyrimidine dehydrogenase (DPD) → 5-6-dihydrofluorouracil DPD is found in liver, intestinal mucosa, tumor and other tissues.
Excretion	Partially metabolized to CO ₂ , which is excreted by the lung Urine, 5% unchanged drug
t _{1/2} plasma	6-20 minutes → Administered by continuous IV infusion Biologic t _{1/2} : prolonged, tissue dependent
CAPECITABINE	Oral prodrug of 5-FU; converted primarily in the liver to inactive intermediate compounds, then to 5-FU in the tumor
Pharmacogenomics	Patients with a complete deficiency of DPD (autosomal recessive trait) are at risk of severe, excessive toxicity. This enzyme deficiency is seen in up to 5% of cancer patients.

Resistance | Therapeutic uses | Toxicities

Resistance	Therapeutic Uses Broad spectrum, solid tumors	Toxicities
Decreased activation - Cancer cells that do not convert 5-FU to 5-FdUMP Altered substrate - Altered or increased thymidylate synthase levels	First line regimen: Metastatic colorectal cancer Synergy: FOLFOX or FOLFIRI Folinic acid—5-FU—thymidylate form ternary complex Topical 5-FU: Actinic keratosis Basal cell carcinoma, superficial Squamous cell carcinoma, in situ Penile carcinoma, in situ Fol: Folinic acid (leucovorin) F: 5-FU OX: Oxaliplatin IRI: Irinotecan	Class toxicities: - Bone marrow suppression, GI, <i>CNS neurotoxicity</i> Hand-foot syndrome - especially with capecitabine - Palmar-plantar erythema: desquamation, pain, and sensitivity to touch Coronary artery vasospasm leading to cardiac ischemia - Angina, MI, arrhythmia, heart failure - Rare, can be fatal.

Cytarabine Mechanisms of Action and Resistance

Mechanisms of Action:

Enters cells via nucleoside transporter →

Cytarabine **deoxycytidine kinase** → Ara-CMP → Ara-CDP → Ara-CTP

1. competitive inhibition of:
 - A. DNA polymerase- α (blocks DNA synthesis), and
 - B. DNA polymerase- β (prevents DNA repair)
2. incorporation into DNA → blocks chain elongation → activation of checkpoint kinases → apoptosis

Mechanisms of Resistance

- ↓ **Conversion to active form:** Loss of deoxycytidine kinase (primary mechanism)
- ↓ **Transport:** Reduced intracellular levels of cytosine analog
- ↑ **Deactivation:** Increased concentration of deactivation enzymes, cytidine deaminase and others may be involved

Cytarabine Therapeutic Uses and Toxicities

Although structurally related, deoxycytidine analogs have different spectrums of action.

Cytarabine

AML: *drug of choice* with daunorubicin or idarubicin

Other leukemias and high-grade lymphomas

Cytarabine has no activity in solid tumors

Toxicities (typical of antimetabolites class)

- Myelosuppression: acute severe leukopenia, thrombocytopenia, and anemia with megaloblastic changes
- Dyspnea, pulmonary infiltrates; can be fatal
- Mucositis; nausea/vomiting; rash; alopecia
- Cerebellar toxicity, high dose or intrathecal (ataxia; slurred speech)
- Cerebral toxicity, high dose or intrathecal (seizures, coma)

Cytarabine
distributes
to the
CNS

Mercaptopurine

Therapeutic Uses

- Acute lymphoblastic leukemia (ALL)
- Component of various regimens

Immunosuppressant

Potent T cell suppression

- Crohn's disease

Toxicities

- Bone marrow suppression
- Hepatotoxicity
- GI symptoms
- Risk of infection

Drug interaction

Allopurinol inhibits xanthine oxidase → shunts 6-MP to formation of more toxic metabolite

🔔 Decrease 6-MP dose to ~25% of usual dose

Check your knowledge

What structural and mechanistic features are common to the antimetabolites class?

How does inhibition of dihydrofolate reductase lead to cell cycle arrest?

Check your knowledge

How does leucovorin reduce bone marrow, liver, and GI toxicity of methotrexate therapy?

What is the mechanism of action for fluorouracil?

Check your knowledge

What is the function of leucovorin in the FOLFOX regimen for colorectal cancer treatment?

What drug is a cytidine analog activated to the triphosphate deoxy form, which inhibits DNA polymerases and causes chain termination, and, with daunorubicin, is a first-line regimen for AML?

Pharmacogenomic poor metabolizers are at increased risk of mercaptopurine toxicities. What is the involved enzyme?

Summary the mechanisms of the antimetabolites

Folate analogs	Block DHFR, thymidylate synthase and early enzymes in purine synthesis blocks RNA and DNA synthesis; cell cycle arrest
Fluoropyrimidines	Incorporation into RNA blocks RNA synthesis; inhibition of thymidylate synthesis inhibits DNA synthesis; cell cycle arrest
Cytidine analogs	Inhibition of DNA polymerase (by cytarabine): Triphosphorylated deoxycytidine analogs compete with dCTP for incorporation into DNA by DNA polymerase, which blocks elongation of growing DNA strand, leading to DNA breaks with activation of p53 and apoptosis if breaks are not repaired. (Other cytidine analogs have slightly different mechanisms.)
Thiopurines	Enzymatic conversion to thio-purines (toxic nucleotides), which are incorporated in DNA and RNA synthesis, leading to cell cycle arrest in the S phase
Cytidine analogs	Activation to triphosphorylated molecule, which inhibits DNA polymerases and terminates strand elongation and triggers apoptosis by incorporation into the growing DNA strand. Other cytidine analogs have additional actions.

Anthracyclines Therapeutic Uses and Resistance mechanisms

Doxorubicin therapeutic uses: Broad spectrum

Wide range of solid tumors, lymphomas, multiple myeloma, sarcomas

Multidrug resistance is common

1. cellular export of drug by MRP transporter family and by the closely related breast cancer resistance protein (BCRP) transporters
2. ↑ glutathione peroxidase activity
3. ↓ activity or mutation of topoisomerase II
4. ↑ repair of DNA strand breaks

MRP: multidrug resistance-associated protein

Anthracyclines Adverse Effects

Cardiotoxicity <i>the most distinctive toxicity</i>	Pericarditis-myocarditis syndrome Chronic: cumulative dose-related, irreversible, progression to cardiomyopathy and acute left ventricular failure <i>Cardiomyopathy is proportional to the cumulative exposure.</i>
Major dose-limiting effects	Myelosuppression (major) Mucositis, stomatitis
Neurotoxicity	Peripheral and CNS
Injection-related	Extravasation / severe local vesicular effects
Radiation recall	Severe local skin reaction in irradiated tissues
Malignancy	Leukemogenic; myelodysplastic syndrome

Vincristine pharmacology

PK	IV Pgp substrate, poor penetration of CSF CYP3A4 metab t _{1/2} 20-45 h <i>Intrathecal admin contraindicated</i>
MOA	Inhibition of microtubule polymerization arresting the cell in metaphase (next slide)
Resistance	MDR resistance is common Multidrug resistant tumor cells display cross-resistance to vinca alkaloids, the epipodophyllotoxins, anthracyclines, and taxanes. • Expression of efflux pumps MDR, MRP, BCRP, Pgp • Mutations in β-tubulin or $\downarrow$ expression of the β-tubulin
Uses	Vinca alkaloids have different activities and therapeutic uses Vincristine: Pediatric and adult leukemias, lymphomas, solid tumors ➤ Treatment of choice to induce remission in childhood leukemia
Vincristine toxicities	Neurotoxicity (main dose-limiting toxicity, can be severe) Peripheral neuropathy, autonomic nervous system dysfunction, motor dysfunction Constipation: can be severe with abdominal pain / obstruction Vesicant Extravasation
	CYP3A4 and P-gp related drug interactions

Taxanes bind β -tubulin with high affinity.

High affinity binding to β -tubulin stabilizes microtubules and prevents microtubule disassembly.



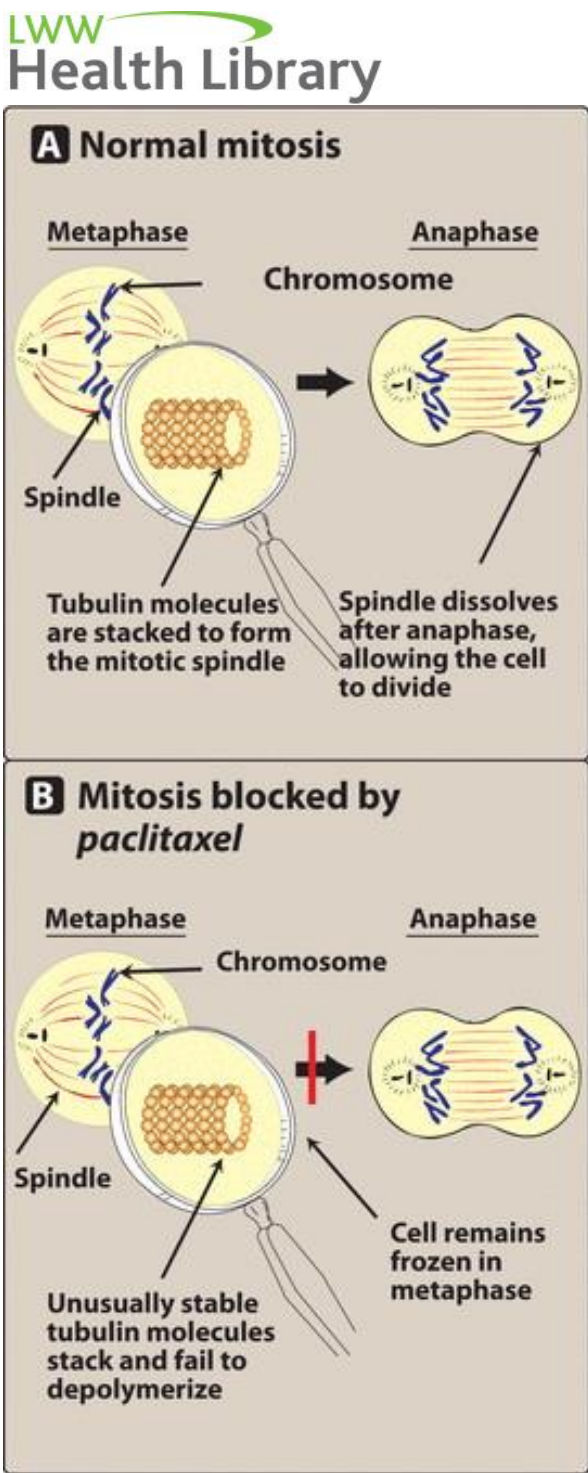
Defects in mitotic spindle assembly, chromosome segregation, and cell division structures derived from microtubules accumulate.



Cell cycle arrest



Apoptosis



Therapeutic Uses

Paclitaxel	Broad spectrum of solid tumors
Nabpaclitaxel	
Docetaxel	
Cabazitaxel	Treatment of castration-resistant metastatic prostate cancer

Resistance:

↑ Expression of MDR genes
Mutations in β -tubulin

Toxicities

- Myelosuppression with neutropenia
- Peripheral neuropathy (“glove-stocking”)
- Radiation recall: Severe reaction in irradiated skin
- Fluid retention, ascites, general edema
- Extravasation / severe vesicular effects
- Administer anthracycline before taxane

Camptothecins

Treatment of metastatic and recurrent
and progressive tumors

Irinotecan

Numerous solid tumors
Metastatic colorectal cancer FOLFIRI
Folinic acid – 5-FU – Irinotecan

Topotecan

Recurrent ovarian cancer
Small cell lung cancer
Cervical cancer
Ewing's sarcoma
Rhabdomyosarcoma
CNS malignancy

Adverse effects

Bone marrow suppression – Dose-limiting: Both

Neutropenia can be severe and potentially fatal
especially when used in combination with other
drugs that cause hematological toxicity

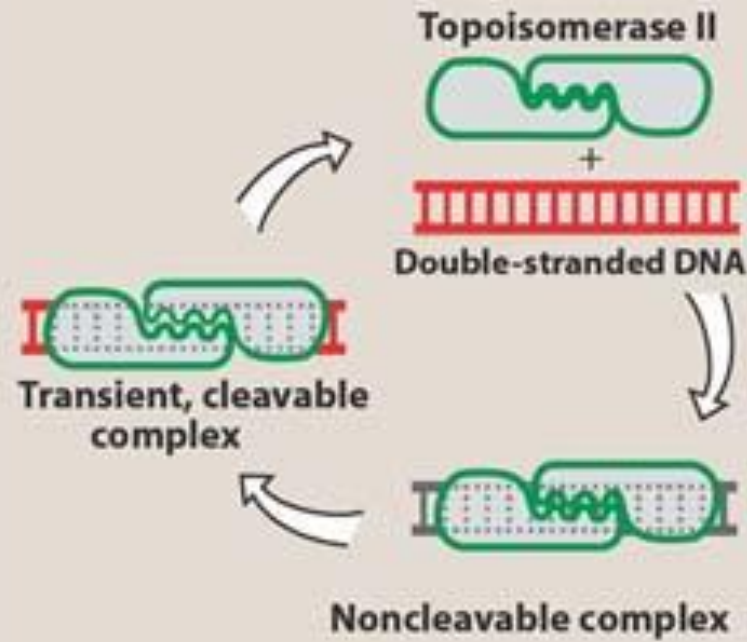
Diarrhea – Dose-limiting: Irinotecan

Diarrhea may be severe with
electrolyte disturbances, dehydration, and
complicated by colitis, ulceration, bleeding
Cholinergic sx may occur w. higher doses

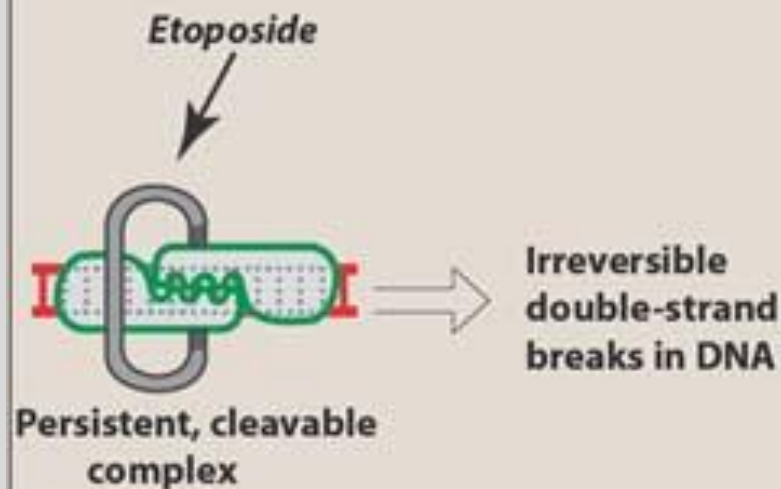
Typical adverse effects of chemo drugs

mucositis, nausea, vomiting, pulmonary toxicity,
renal toxicity, etc.

A Normal catalytic cycle
of topoisomerase



B Etoposide leads to double-strand breaks in DNA



Epipodophyllotoxins Mechanisms

1. Formation of a ternary complex
drug-DNA-topoisomerase II complex
2. Inhibition of topoisomerase II → prevents resealing of double-strand breaks
3. The enzyme remains bound to the free end of the broken DNA strand → accumulation of DNA breaks → arrest → apoptosis

Cell cycle arrest in late S and G2 phase

Resistance Mechanisms

1. Efflux:
2. Topoisomerase II alterations
2. Mutations in *p53* tumor suppressor gene
 - P-53 is required for inducing apoptosis

Check your knowledge

A broad spectrum chemotherapy causes irreversible cardiotoxicity that can lead to left ventricular dysfunction, cardiomyopathy, and heart failure, apparently resulting from generation of free radicals and lipid peroxidation. What is the mechanism of action of this drug?

What are the other dose-limiting toxicities of this drug?

A patient with a history of rhabdosarcoma treated with a chemotherapy regimen that includes vincristine continues to suffer with painful pins and needles sensation, motor weakness and muscle cramps in his upper and lower extremities. What is the adverse effect?

Check your knowledge

What is the difference between the mechanisms of action of the vinca alkaloids and the taxanes?

What are the dose-limiting toxicities of each drug in the regimen?

Check your knowledge

A patient with KRAS mutated metastatic colorectal cancer is given the FOLFIRI regimen. What does “IRI” represent and what is its dose-limiting side effect?

What are other dose-limiting toxicities of this drug?

A 28-year-old male diagnosed with testicular cancer is treated with the first-line regimen, BEP regimen. E represents a drug that prevents resealing of the double strand breaks in DNA ultimately leading to cell cycle arrest in S-G2 phase. What is the name of the drug?

Check your knowledge

What are the mechanisms of action of the other two drugs in the BEP regimen?

(I don't expect you to know the answer so I will tell you because it is Boards Relevant.)

What is an important mechanism of resistance common to the anthracyclines and the antitumor natural products?

Class / Drugs	Mechanism
Alkylating/ Adducting Agents: Cell cycle nonspecific (occurs at any stage). Cells are most vulnerable in G1 and S phases	
Alkylating agents *Cyclophosphamide Ifosfamide Many others	Transfer alkyl groups N7 position of guanine (mainly) and other points on DNA. Also react with other cellular nucleophiles. Alkylation of DNA results inter- and intrastrand cross-linking, abnormal base pairing (miscoding), scission of DNA backbone, and strand breakage.
Platinum analogs *Cisplatin *Carboplatin *Oxaliplatin	Binds guanine in DNA, forming adducts with inter- and intrastrand cross-linking, which inhibits polymerases, thus inhibiting DNA replication/synthesis and RNA synthesis.

Class / Drugs	Mechanism
Antimetabolites: S phase cell cycle specific activity	
Antifolates: *Methotrexate Two others	Inhibit DHFR and thymidylate synthase, which inhibits de novo purine nucleotide synthesis
Fluoropyrimidines: *5-Fluorouracil *Capecitabine One more	1- 5-FU is converted to 5-FdUMP. Forms stable complex: 5-FdUMP-folate-thymidylate synthase that inhibits thymidylate synthesis 2- Incorporation of FdUTP into DNA leads to strand breaks and faulty DNA replication 3- Incorporation of FUTP in RNA disrupts RNA production and function
Deoxycytidine analogs: *Cytarabine Gemcitabine	Cyt: Converted to active Ara-CTP that inhibits DNA polymerase α , which blocks DNA synthesis, and polymerase β , which blocks DNA repair; AraCTP incorporation into DNA and RNA inhibits their synthesis. Gem: similar, plus inhibition of ribonucleotide reductase
Purine antagonists *6-Mercaptopurine 6-Thioguanine Two others	Converted to toxic purine nucleotides that are incorporated into DNA and RNA inhibiting their synthesis

Antitumor Antibiotics: Cell cycle nonspecific	
Antitumor Antibiotics: *Doxorubicin Four other anthracyclines, Two other antibiotics	1- Topoisomerase II inhibition and irreversible DNA strand scission 2- Intercalation by high-affinity binding to DNA, blocking DNA and RNA synthesis 3- Generation of semiquinone free radicals that cause damage to DNA, proteins, and cell membranes
Microtubule Inhibitors: M phase cell cycle specific in metaphase	
Vinca alkaloids: Vinblastine *Vincristine Vinorelbine	Bind β -tubulin → blocks tubulin polymerization into microtubules → disrupts formation of mitotic spindle → chromosomal segregation and cell proliferation are inhibited
Taxanes: *Paclitaxel *Docetaxel Two others	Bind with high-affinity to microtubules → stabilizes existing microtubules → prevents mitotic spindle function and cell division → cell cycle arrest → apoptosis
Topoisomerase Inhibitors: S phase cell cycle specific	
Camptothecins: *Irinotecan Topotecan	Inhibit topoisomerase I, the enzyme that cuts and religates single DNA strands
Epipodophyllotoxins: *Etoposide	Inhibits topoisomerase II, the enzyme that cuts and re-ligates double stranded DNA

Check your knowledge

What class of drugs induces cell death by nucleophilic attack on DNA bases, esp. the N7 guanine, leading to inter- and intrastrand cross-linking, abnormal base pairing, miscoding, and strand breakage.

- The alkylating agents. DNA damage activates of DNA repair systems that create DNA strand breaks by repair enzymes, which activates the apoptosis pathway.

What cellular mechanisms are essential for cell death?

- Recognition of the adduct
- Intact mismatch repair and DNA damage repair systems to induce DNA strand breaks by repair enzymes
- Normal p53 gene and activation of checkpoint proteins
- **Resistance develops rapidly when a drug is used as a single agent**

What is the active cytotoxic metabolite of cyclophosphamide (and ifosfamide)?

- Phosphoramidate mustard: cyclophosphamide → hydroxycyclophosphamide → aldophosphamide → phosphoramidate mustard

Check your knowledge

What is the name of the urotoxic metabolite and how can the toxicity be mitigated?

- Acrolein toxicity to bladder epithelium causes hemorrhagic cystitis.
- Mesna conjugates and inactivates acrolein, decreasing the risk of hemorrhagic cystitis

What is a mechanisms of tumor resistance to cyclophosphamide and ifosphamide?

- Increased expression of aldehyde dehydrogenase increases the rate of metabolism of aldophosphamide to inactive carboxyphosphamide.
- Increased expression glutathione transferase detoxifies alkylating intermediates

Cyclophosphamide is used in the treatment of what categories of cancers?

- Numerous solid and blood tumors and immunosuppression in autoimmune diseases.

List the common and serious toxicities of cyclophosphamide.

- bone marrow suppression, liver tox., cardiotox, lung tox, malignancies and class tox.

Check your knowledge

How does the mechanism of the platinum analogs compare with the mechanism of the alkylating agents?

- Alkylating agents: Nucleophilic attack on DNA → cross-linking → activate DNA repair mechanisms → apoptosis
- Platinum analogs: Diffuse through copper transporter → aquation to positively charged molecules → nucleophilic attack on DNA → cross-linking → activate DNA repair mechanisms → apoptosis. Platinum analogs also bind nuclear and cytoplasmic proteins
- Apoptosis requires that p53 and other checkpoint proteins are intact.

What are the main mechanisms of resistance to platinum analogs?

- Efflux (ATP7A/B copper exporters)
- Inactivation (metalloproteinases, glutathione-S-transferase, sulfhydryls)
- Loss of DNA mismatch repair and decreased apoptosis, and other mechanisms.

Check your knowledge

What is the definition of dose-limiting toxicities?

- Specific adverse effects of a drug that are severe enough to prevent an increase in the dose. They are critical in determining the maximum tolerated dose during clinical trials, especially in oncology.

What are the dose-limiting toxicities of cisplatin, carboplatin, and oxaliplatin?

- Cisplatin: nephrotoxicity; carboplatin: bone marrow suppression; oxaliplatin: peripheral neuropathy, acute and delayed (progressive)

In sequential administration of a platinum analog with paclitaxel or docetaxel, why is the taxane administered first?

- To limit myelosuppression (taxane) and increase efficacy.
- Pharmacokinetic data during clinical trials with paclitaxel administered in sequential infusion with cisplatin demonstrated a decrease in paclitaxel clearance when paclitaxel was administered after cisplatin.
- The mechanism has not been identified.
- Administer paclitaxel before cisplatin, when given as sequential infusions, to limit toxicity. Problems associated with other taxane/platinum combinations are possible, although unsubstantiated. Administering the taxane derivative before the platinum derivative seems prudent.

Check your knowledge

A broad spectrum chemotherapy causes irreversible cardiotoxicity that can lead to left ventricular dysfunction, cardiomyopathy, and heart failure, apparently resulting from generation of free radicals and lipid peroxidation. What is the mechanism of action of this drug?

- 1) DNA intercalation, 2) inhibition of topoisomerase II which inhibits DNA and RNA synthesis, DNA repair, and results in DNA fragmentation, 3) generation of free radicals, which damage DNA, membrane functions.

What are the other dose-limiting toxicities of this drug?

- Bone marrow suppression, mucositis/stomatitis

A patient with a history of rhabdosarcoma treated with a chemotherapy regimen that includes vincristine continues to suffer with painful pins and needles sensation, motor weakness and muscle cramps in his upper and lower extremities. What is the adverse effect?

- Sensorimotor peripheral neuropathy, which is a dose-limiting toxicity of vincristine.

Check your knowledge

What is the difference between the mechanisms of action of the vinca alkaloids and the taxanes?

- Vinca alkaloids inhibit polymerization in microtubules
- Taxanes stabilize microtubules and prevent depolymerization
- Both suppress spindle microtubule function → cell cycle arrest → apoptosis

A patient diagnosed with breast cancer receives paclitaxel intravenous infusion once weekly every three weeks as a component of her adjuvant chemotherapy regimen that includes doxorubicin and cyclophosphamide.

What are the dose-limiting toxicities of each drug in the regimen?

- Paclitaxel: bone marrow suppression and peripheral neuropathy (sensorimotor, glove-stocking)
- Doxorubicin: cardiotoxicity/cardiomyopathy (cumulative dose); bone marrow suppression
- Cyclophosphamide: bone marrow suppression and hemorrhagic cystitis (due to acrolein metabolites), congestive heart failure

Check your knowledge

A patient with KRAS mutated metastatic colorectal cancer is given the FOLFIRI regimen. What does “IRI” represent and what is its dose-limiting side effect?

- IRI = irinotecan, a prodrug converted to toxic SN-38, a topoisomerase I inhibitor prevents resealing of DNA strand, leading to strand breaks and cell cycle arrest in the S phase.
- DNA strand breaks causes cell cycle arrest in S phase.

What are other dose-limiting toxicities of this drug?

- Bone marrow suppression, mucositis/stomatitis

A 28-year-old male diagnosed with testicular cancer is treated with the first-line regimen, BEP regimen. E represents a drug that prevents resealing of the double strand breaks in DNA ultimately leading to cell cycle arrest in S-G2 phase. What is the name of the drug?

- Etoposide inhibits of topo II by forming drug-DNA-topoisomerase II complex, which prevents resealing of double-strand breaks, causing irreversible double-strand breaks.
- Accumulation of DNA double strand breaks causes cell cycle arrest in late S and G2 phases and apoptosis.

Check your knowledge

What are the mechanisms of action of the other two drugs in the BEP regimen?

(I don't expect you to know the answer so I will tell you because it is Boards Relevant.)

- B=Bleomycin, an antitumor antibiotic that forms a complex with DNA and iron: DNA-B-Fe²⁺, which is oxidized to DNA-B-Fe³⁺. Superoxide and hydroxyl radicals are formed, which attack phosphodiesterase bonds of DNA, resulting in strand breaks and chromosomal aberrations. Bleomycin causes pulmonary fibrosis (“bleomycin lung”).
- P=platinum=cisplatin. Chlorine molecules on the structure are replaced by water molecules (aquation) in the low-chloride environment inside the tumor cell, which then attacks purine residues on DNA, especially the N⁷ of guanine, leading to inter- and intrastrand breaks and miscoding – apoptosis (if recognized by p53 and other checkpoint proteins). Nephrotoxicity is the dose-limiting toxicity.

What is an important mechanism of resistance common to the anthracyclines and the antitumor natural products?

- Efflux via upregulated expression of efflux pumps: multidrug resistance (MDR) pumps including MDR1= P-glycoprotein, multidrug resistance-associated protein (MRP), breast cancer resistance pump (BCRP).